

RESEARCH ARTICLE

Enhancing Efficient Personalized Learning and Educational Management in Universities Using Graph Neural Networks in Intelligent Tutoring Systems

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ABSTRACT Personalized learning has become an essential strategy for accommodating the diverse and evolving needs of learners within contemporary educational environments. Traditional educational models, however, often fall short in capturing the intricate and ever-changing nature of individual learning trajectories. In response to these limitations, this study introduces an innovative framework that utilizes graph-based modeling techniques to advance the field of personalized learning black and university education management. Central to this framework are two core components: the Knowledge Dynamics Model (KDM) and the Adaptive Knowledge Reinforcement Strategy (AKRS). These components collaboratively facilitate the modeling of complex knowledge states, individual learning paces, and the phenomenon of cognitive decay over time. By representing curricular content as a directed acyclic graph (DAG), the framework accurately depicts learning sequences, prerequisite structures, and conceptual dependencies, thus enabling tailored instructional pathways and enhanced skill transfer in university education management scenarios. Moreover, the approach incorporates real-time learner performance feedback and adaptive reinforcement mechanisms to promote sustained knowledge retention and mastery. Empirical evaluations confirm that the integration of KDM and AKRS leads to improved learning outcomes, including a 15.3% increase in learner engagement, by dynamically aligning instructional methods with learners' cognitive development. This graph-based, data-driven framework offers a responsive and effective approach to delivering personalized education that evolves alongside each learner's progress.

INDEX TERMS University education management, personalized learning, knowledge dynamics, adaptive learning, graph-based modeling, cognitive reinforcement.

I. INTRODUCTION

Enhancing personalized learning in intelligent tutoring systems (ITS) has become an increasingly critical research area, driven by the need to provide more effective, tailored educational experiences [1]. Traditional ITS approaches often relied on rule-based systems and knowledge representation techniques to model student understanding and deliver personalized instruction [2]. These methods, however, struggled

to capture the complexity and variability of human learning, limiting their effectiveness in diverse real-world contexts. With the advent of data-driven methods, researchers began integrating machine learning (ML) approaches, enabling systems to adapt more dynamically to student needs [3].

Education is a critical foundation for personal development and societal progress, encompassing the acquisition of knowledge, skills, values, and competencies essential for both individual growth and societal advancement. Education can be broadly categorized into formal, non-formal, and informal learning contexts. Formal education typically refers to

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structured learning provided by schools, colleges, and universities with well-defined curricula and assessment standards, while non-formal education includes organized learning activities outside formal institutions, such as vocational training and adult education. Informal education captures unstructured, experiential learning that occurs throughout daily life. Contemporary educational systems face multiple challenges, including unequal access to quality education, growing demands for personalization, and the need to align learning outcomes with rapidly evolving socio-economic requirements. Disparities in educational resources, digital divides, and differences in teacher preparedness remain significant obstacles to achieving equitable and effective education. At the same time, the increasing integration of digital platforms, artificial intelligence, and immersive technologies has fundamentally transformed how educational content is delivered and consumed. While these technologies create new opportunities for personalized and scalable learning, their effective adoption requires careful consideration of pedagogical design, digital literacy, and infrastructure constraints. Against this backdrop, emerging educational paradigms such as personalized learning, competency-based education, and lifelong learning pathways have gained increasing attention. These models emphasize adaptive instruction, continuous assessment, and learner-centered progression, highlighting the need for intelligent systems that can model learning dynamics and support data-driven educational decision-making. To effectively respond to these evolving educational challenges, it is essential to develop intelligent systems that are not only grounded in pedagogical theory but also capable of dynamic, data-driven adaptation. The increasing availability of multimodal learning data and advancements in deep learning, particularly graph-based models, provide new opportunities for modeling complex educational relationships and personalizing instruction at scale. In this context, we propose a graph neural network–based framework that integrates cognitive modeling with adaptive feedback mechanisms to support both individual learner progression and institutional decision-making. Yet, these methods often lacked the ability to fully utilize the rich, structured relationships present in educational data. This is particularly relevant in university education management, where personalized support and academic decision-making require fine-grained understanding of student performance and curriculum interdependencies.

More recently, the rise of deep learning and graph neural networks (GNNs) has provided a promising path forward, offering the capacity to model complex interactions between students, educational content, and pedagogical strategies at scale [4]. This shift not only addresses the limitations of prior approaches but also enables the development of more responsive, context-aware learning environments, making personalized education more precise and impactful. Early efforts in intelligent tutoring systems primarily emphasized formalized instructional design and domain modeling. Researchers constructed systems that relied on structured,

rule-governed frameworks to simulate learning processes and assess student progress [5]. These models offered clarity and interpretability, and they enabled the encoding of expert knowledge in controlled ways. However, the rigidity of such systems limited their ability to accommodate the nuanced, evolving nature of individual learners [6]. As a result, their scalability to broader educational scenarios and learner diversity was constrained, and maintaining their underlying knowledge structures proved labor-intensive [7].

As the availability of educational interaction data increased, researchers began exploring algorithmic approaches that could infer learner patterns from empirical observations [8]. These models incorporated statistical techniques to personalize content delivery based on observed student behavior. By learning from historical data, such systems could dynamically adjust instructional strategies and provide targeted support [9]. However, many of these models simplified the learning process into isolated or sequential tasks, failing to capture the web of interdependencies among educational concepts [10]. This abstraction limited their ability to generalize across subjects and learners, reducing their impact in open-ended learning environments [11].

Still, these approaches advanced the field by introducing adaptability and predictive capabilities, setting the stage for more nuanced personalization. Recent developments have emphasized integrated approaches that draw on deep learning and structural modeling to capture the multifaceted nature of education [12]. In particular, techniques that model relationships among students, concepts, and learning activities as interconnected systems have gained attention [13]. These methods excel at extracting latent structures from complex datasets and can adaptively reflect the evolving understanding of individual learners [14]. Through mechanisms that consider both immediate interactions and broader learning trajectories, such models provide a richer representation of educational progress [15]. While these innovations come with practical challenges—such as computational demands and data requirements—their potential to deliver highly personalized, scalable, and context-sensitive tutoring makes them a promising direction in ITS research. In the context of university education management, these innovations offer a pathway to bridge the gap between individual learning behaviors and institutional-level optimization. Administrators and educators can leverage such systems to monitor academic progress across cohorts, detect early signs of disengagement, and implement targeted interventions aligned with strategic teaching goals.

Given the limitations of prior approaches, our method aims to overcome these challenges by integrating GNNs into ITS to enhance personalization and adaptivity. By leveraging the powerful relational modeling capabilities of GNNs, our approach can capture the nuanced, context-specific relationships between students, learning materials, and instructional strategies. This method addresses the scalability issues of symbolic AI, the data fragmentation of traditional ML, and

the context insensitivity of earlier deep learning approaches, providing a more comprehensive and flexible framework for personalized learning.

Intelligent Tutoring Systems (ITS) have evolved from early rule-based and expert-driven approaches to modern, data-driven paradigms. While traditional ITS offered structured instruction and clear interpretability, they lacked scalability and adaptability to diverse learner behaviors. The emergence of Graph Neural Networks (GNNs) has opened new opportunities for modeling the complex, interconnected nature of student knowledge, but current GNN-based ITS still face significant challenges, including limited scalability to large and heterogeneous learning environments, difficulty in interpreting model decisions, and insufficient mechanisms for adaptive reinforcement of knowledge gaps. To tackle these challenges, we introduce a unified framework that combines the Knowledge Dynamics Model (KDM) with the Adaptive Knowledge Reinforcement Strategy (AKRS). KDM leverages GNNs to represent and update knowledge states as a dynamic directed acyclic graph (DAG), capturing intricate dependencies among learning units. AKRS adaptively reinforces weaker concepts based on real-time performance feedback, improving both learning efficiency and interpretability by directly linking system recommendations to student outcomes. Together, these components fill a critical gap in current GNN-based ITS by enabling scalable, interpretable, and performance-driven personalization in educational settings. Our framework can be intuitively understood as a personalized learning navigator. The Knowledge Dynamics Model (KDM) tracks what a learner knows and forgets over time, while the Adaptive Knowledge Reinforcement Strategy (AKRS) continuously recommends what to study next, like a GPS recalculating routes based on real-time traffic.

A. CONTRIBUTIONS

This work makes the following key contributions to the field of intelligent tutoring systems and graph-based personalized learning:

- We propose a unified GNN-based framework that combines a Knowledge Dynamics Model (KDM) with an Adaptive Knowledge Reinforcement Strategy (AKRS), enabling personalized learning through real-time modeling and dynamic path optimization.
- We introduce a competency-aware feedback loop that incorporates a cross-domain mapping matrix, allowing the system to identify latent skill dependencies and reinforce learning across related knowledge areas.
- Our framework integrates cognitive modeling principles such as forgetting curves and learning rate adaptation directly into the graph learning process, bridging theoretical learning science with practical algorithm design.
- We demonstrate superior performance over existing GNN-ITS models through extensive evaluation on

five diverse educational datasets, and we validate the effectiveness of each module via detailed ablation studies.

- We enhance deployment feasibility by supporting privacy-preserving federated learning and lightweight inference for resource-constrained environments.

II. RELATED WORK

A. GRAPH NEURAL NETWORKS FOR STUDENT KNOWLEDGE MODELING

Graph Neural Networks (GNNs) have emerged as a powerful tool for representing and analyzing complex relational data, making them particularly well-suited for student knowledge modeling in Intelligent Tutoring Systems (ITS) [16]. Traditional approaches to student modeling, such as Bayesian Knowledge Tracing (BKT) and Deep Knowledge Tracing (DKT), primarily rely on sequential data, treating the learning process as a series of independent interactions between a student and an educational system. Nevertheless, such approaches frequently fall short in representing the complex relationships among various knowledge units, cognitive abilities, and educational constructs [17]. GNNs address this limitation by representing the relationships between knowledge components as a graph, where nodes correspond to individual skills or concepts, and edges represent their semantic, pedagogical, or cognitive connections. This graph-based structure allows GNNs to capture both direct and indirect relationships between concepts, reflecting the hierarchical and interconnected nature of human knowledge [18]. For example, in mathematics, understanding calculus inherently requires prior mastery of algebra and trigonometry, creating a layered dependency structure that GNNs can effectively model. Moreover, graph neural networks possess the ability to continuously refine the status of individual nodes as learners engage in varying instructional tasks, thereby enabling a more accurate modeling of their shifting knowledge profiles [19]. This functionality proves especially beneficial in the realm of individualized education, where the objective is to tailor instructional material according to each learner's distinct capabilities and areas for improvement. Recent studies have demonstrated that GNN-based approaches, such as Knowledge Tracing GNN (KT-GNN) and Graph-based Cognitive Diagnosis (GCD), outperform traditional methods by leveraging this structural information, resulting in more accurate predictions of student performance and more effective personalized learning recommendations. Moreover, the ability of GNNs to incorporate heterogeneous data types, including textual explanations, video content, and interactive exercises, further enhances their applicability in modern ITS [20]. This integration of multimodal data allows GNNs to provide a richer, more context-aware understanding of student knowledge, improving both engagement and learning outcomes. However, many of these GNN-based models treat knowledge as static graphs or optimize only for prediction accuracy, often neglecting the temporal evolution

of learning, cognitive variability, and reinforcement needs. Few models explicitly incorporate knowledge forgetting, adaptive spacing, or concept-level reinforcement into the graph structure, which limits their effectiveness in modeling real-world, evolving student behavior. In contrast, our framework introduces a Knowledge Dynamics Model (KDM) that models knowledge as a dynamic DAG, enabling real-time updates that reflect acquisition and decay, and an Adaptive Knowledge Reinforcement Strategy (AKRS) that continuously calibrates learning interventions based on performance and cognitive load.

B. KNOWLEDGE TRACING WITH GRAPH-BASED MODELS

Knowledge tracing (KT) is a central task in personalized education, aimed at modeling students' mastery of concepts over time. Early deep learning approaches such as Deep Knowledge Tracing (DKT) introduced recurrent neural networks to capture temporal dependencies in student interactions [21], offering significant improvements over traditional Bayesian methods [22]. Building on this, the Dynamic Key-Value Memory Network (DKVMN) introduced an external memory structure to separate concept representations from student mastery, enhancing interpretability and scalability [23]. More recently, graph-based models such as GIKT have incorporated relational structures between questions and concepts to better capture the underlying learning dynamics [24]. However, these models primarily focus on prediction accuracy rather than cognitive control or curriculum adaptability. In contrast, our proposed KDM-AKRS framework extends beyond prediction to support dynamic reinforcement, cognitive load modeling, and adaptive curriculum path optimization, thus offering a more holistic solution for intelligent tutoring in higher education environments. These methods largely prioritize predictive modeling without accounting for long-term knowledge retention, dynamic reinforcement, or cross-domain reasoning. For example, while DKVMN improves interpretability, it lacks mechanisms for real-time adaptation or cross-concept reinforcement. Our KDM-AKRS framework directly addresses these gaps by embedding forgetting functions, adaptive learning rate matrices, and curriculum-sensitive feedback loops into the model, allowing it to both predict performance and proactively guide learning in dynamic, higher-education contexts.

C. PERSONALIZATION AND ADAPTIVE LEARNING USING GRAPH NEURAL NETWORKS

Personalization is at the heart of effective intelligent tutoring systems, and Graph Neural Networks (GNNs) offer a powerful framework for achieving this by modeling complex, personalized learning pathways [25]. Traditional adaptive learning systems often rely on predefined decision rules or shallow machine learning models that fail to capture the intricate, non-linear relationships inherent in educational data. GNNs, by contrast, leverage the graph structure to represent the interconnected nature of educational content

and student interactions, enabling a more nuanced understanding of individual learning trajectories [26]. In this context, each student can be represented as a dynamic subgraph within the larger knowledge graph, where nodes correspond to the concepts a student has encountered, and edges capture the cognitive relationships and prerequisite dependencies between these concepts. This approach allows GNNs to model the influence of prior knowledge on current learning, capture the spread of learning effects across related topics, and identify potential misconceptions or learning gaps that might hinder progress [27]. For example, if a student struggles with quadratic equations, a GNN-based ITS can infer potential weaknesses in foundational algebra concepts and adapt the learning path accordingly. This personalized approach not only improves learning outcomes by targeting specific areas of need but also enhances student motivation by providing appropriately challenging content. Furthermore, GNNs can incorporate real-time performance data, such as quiz scores, interaction logs, and response times, to continuously refine their understanding of each student's knowledge state [28]. This real-time feedback loop enables the system to dynamically adjust instructional strategies, such as offering additional practice problems, providing personalized hints, or adjusting the difficulty level of subsequent exercises. However, these personalization strategies often rely on static student profiles or shallow updates that fail to capture the nuanced, time-dependent nature of learning. Most existing systems do not provide interpretable pathways for educators to intervene or understand the rationale behind content sequencing. Our approach enhances these capabilities by modeling knowledge as a dynamic, interpretable graph and aligning content delivery with inferred cognitive states, enabling scalable yet personalized learning trajectories grounded in pedagogical theory.

D. ASSESSMENT AND FEEDBACK OPTIMIZATION WITH GNNs IN ITS

Effective assessment and feedback are critical components of personalized learning, providing students with timely insights into their strengths and areas for improvement [29]. Graph Neural Networks (GNNs) are particularly well-suited for this task, as they can model the complex, context-dependent nature of student performance within a knowledge graph. Unlike conventional assessment methods, which often rely on fixed, one-size-fits-all test structures, GNNs can dynamically generate personalized assessments based on each student's current knowledge state [30]. This is achieved by leveraging the graph's structure to identify key concepts that are both relevant to the current learning goals and closely related to the student's past performance. For instance, if a student has recently demonstrated mastery in algebra but struggles with calculus, a GNN-based assessment system can prioritize questions that bridge these two areas, reinforcing foundational knowledge while introducing more advanced concepts [31]. Furthermore, GNNs can analyze

response patterns to detect common misconceptions, cognitive bottlenecks, and knowledge gaps that might not be apparent through traditional assessment techniques. This granular understanding of student learning enables ITS to provide more targeted and actionable feedback, such as recommending specific practice problems, offering tailored hints, or suggesting relevant instructional videos [32]. GNNs can incorporate multimodal data, including clickstream logs, eye-tracking data, and even emotional cues, to assess student engagement and motivation, providing a more comprehensive view of learning progress. Yet, the majority of these systems lack continuous feedback adaptation and rely on heuristics or static assessment pathways. Furthermore, existing GNN-based assessment systems do not explicitly connect performance evaluation with reinforcement planning. Our AKRS component directly addresses this by linking performance gaps with real-time path adjustments, allowing assessments not only to diagnose, but also to drive immediate, individualized instructional changes.

E. SUMMARY AND POSITIONING

In summary, while recent GNN-based ITS models have made significant strides in predictive accuracy and personalization, they often fall short in integrating cognitive dynamics, real-time reinforcement, and educational interpretability. Our proposed KDM-AKRS framework bridges this gap by modeling learning as a dynamic process influenced by prior knowledge, performance feedback, and cognitive variability. It combines the structural advantages of graph modeling with pedagogical insights from learning science to deliver a unified, adaptive, and explainable tutoring system. This positions our work as a novel and necessary extension of current state-of-the-art ITS research.

III. METHOD

A. OVERVIEW

Building upon the educational motivations outlined earlier, our method aims to operationalize key instructional principles—such as personalization, mastery learning, and cognitive regulation—through a unified graph-based framework. By translating these pedagogical goals into formal computational structures, the proposed system enables scalable and interpretable personalization, directly addressing the limitations of traditional intelligent tutoring systems.

As illustrated in Figure 1, the overall architecture consists of two tightly coupled components: the Knowledge Dynamics Model (KDM) and the Adaptive Knowledge Reinforcement Strategy (AKRS). KDM is responsible for representing and updating each learner's knowledge state in the form of a dynamic directed acyclic graph (DAG). This graph encodes prerequisite relationships among learning concepts, and evolves over time to reflect learning gains, forgetting effects, and performance-based adjustments.

AKRS builds upon the updated knowledge state produced by KDM and focuses on adaptive decision-making.

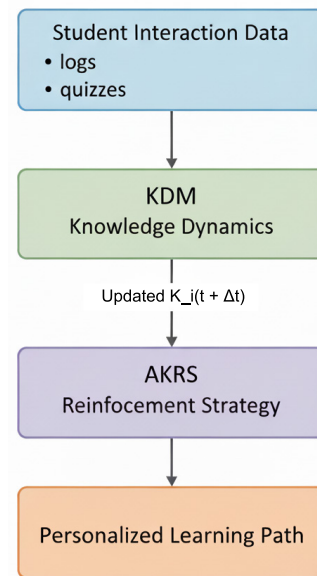


FIGURE 1. Interaction between the Knowledge Dynamics Model (KDM) and the Adaptive Knowledge Reinforcement Strategy (AKRS). KDM provides real-time knowledge state estimation, while AKRS uses this information to guide personalized reinforcement and learning path optimization.

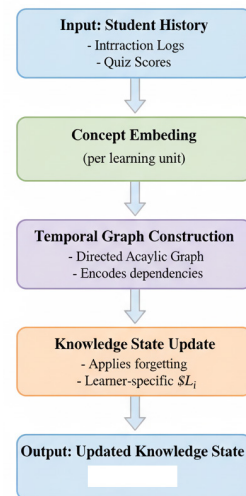
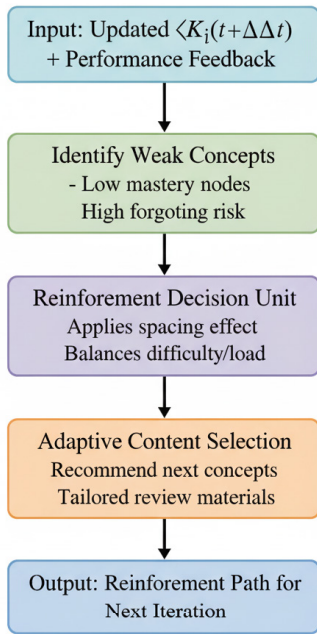


FIGURE 2. Workflow of the Knowledge Dynamics Model (KDM). KDM models the temporal evolution of student knowledge by constructing a dynamic directed acyclic graph from interaction data and continuously updating concept mastery states with consideration of learning and forgetting effects.

It identifies weak or unstable knowledge components, determines reinforcement priorities, and schedules future learning activities while considering cognitive load and concept interdependence. This process forms a closed-loop architecture in which learner performance dynamically informs both knowledge modeling and instructional path planning.

Figure 2 details the internal workflow of the KDM module, showing how historical interaction data is embedded,



Adaptive Content Reinforcement for Intelligent Tutoring Systems

FIGURE 3. Workflow of the Adaptive Knowledge Reinforcement Strategy (AKRS). AKRS identifies weak or unstable knowledge components from updated knowledge states and adaptively generates reinforcement paths by balancing learning effectiveness and cognitive load.

structured into a concept graph, and updated based on student performance and forgetting behavior. Figure 3 presents the operational flow of the AKRS module, highlighting how it transforms knowledge state insights into reinforcement strategies that drive personalized content selection.

This overview sets the methodological foundation for the remainder of the paper. Section III-B introduces the formal definitions and mathematical preliminaries. Section III-C elaborates on the structure and update rules of the Knowledge Dynamics Model. Section III-D describes the design and operational logic of the Adaptive Knowledge Reinforcement Strategy. Together, these components define a closed-loop, interpretable, and cognitively informed approach to personalized learning based on graph neural networks.

The design of our proposed framework is informed by well-established educational theories that support effective learning. The Knowledge Dynamics Model (KDM) implicitly aligns with Bloom's taxonomy by enabling differentiated tracking of student progress across cognitive levels—such as understanding foundational concepts, applying problem-solving techniques, and analyzing multi-step procedures. Each concept node within the graph can be tagged with a cognitive level, allowing the system to sequence learning experiences that gradually progress from lower-order to higher-order thinking skills. The Adaptive Knowledge Reinforcement Strategy (AKRS) is guided by principles from cognitive load theory. By identifying weak knowledge areas and spacing reinforcement activities over time, AKRS aims

to optimize the learner's intrinsic cognitive load while minimizing extraneous load. This dynamic scheduling reduces overload from redundant content while maintaining engagement through appropriately challenging material. AKRS supports core mechanisms of self-regulated learning (SRL) by enabling learners to receive timely, data-driven feedback, monitor their progression through the curriculum, and adapt their learning pace based on personalized reinforcement paths. These theoretical alignments ensure that the technical mechanisms embedded in our framework are pedagogically grounded, facilitating not just prediction and adaptation but also meaningful learning aligned with instructional design principles.

B. PRELIMINARIES

Education is a process of knowledge acquisition and skill development, often involving structured sequences of instruction. To formally model this process, we begin by defining key components that represent learners, content structure, and cognitive dynamics. Let $S = \{s_1, s_2, \dots, s_N\}$ denote a set of students, and $C = \{c_1, c_2, \dots, c_M\}$ a set of courses or knowledge units. Each student s_i possesses a knowledge state vector $K_i(t) \in \mathbb{R}^d$, where each dimension corresponds to a specific concept (e.g., algebra, logic, calculus). This state evolves over time as learning occurs:

$$K_i(t + \Delta t) = K_i(t) + L_i F(K_i(t), I_i(t), R_i, C_i) - D(K_i(t), t) \quad (1)$$

For consistency throughout the paper, we adopt a unified formulation of the knowledge state update. Here, L_i denotes the personalized learning rate matrix for student i , $F(\cdot)$ is the learning gain function based on instructional input, resources, and cognitive context, and $D(\cdot)$ represents the knowledge decay function. This structure is used consistently in all subsequent equations.

To reflect prerequisite constraints and course sequencing, we model the curriculum as a Directed Acyclic Graph (DAG):

$$\mathcal{P} = (C, \mathcal{E}), \quad \mathcal{E} \subseteq C \times C, \quad (2)$$

An edge $(c_i, c_j) \in \mathcal{E}$ implies that course c_i is a prerequisite for course c_j . For example, Linear Algebra may be required before Machine Learning. This structure encodes the logical order in which knowledge should be acquired. Learning is also subject to forgetting. Without review or reinforcement, knowledge fades over time. This phenomenon is modeled using exponential decay:

$$R(t) = K_0 e^{-\lambda t}, \quad (3)$$

where K_0 is the initial knowledge strength and λ is the forgetting rate. This mirrors real-world memory behavior observed in cognitive science, such as the Ebbinghaus forgetting curve. Performance assessment is central to adaptive learning. We denote a student's score at time t as $P_i(t)$, calculated based on their current knowledge vector and the relevance of test items. For simplicity, this can be

viewed as a weighted sum over mastered concepts. To close the loop, a feedback mechanism compares the learner's performance against a desired target T_i . The resulting error drives knowledge reinforcement in future steps:

$$\Delta K_i(t) = \alpha(T_i - P_i(t)), \quad (4)$$

This dynamic ensures that weak areas receive more attention. For instance, if a student scores below their target on Calculus, the system might schedule more review in related foundational topics like Algebra. Together, these core concepts establish the mathematical backbone of our framework. They serve as the basis for modeling knowledge dynamics, course structures, and personalized reinforcement strategies described in subsequent sections.

To enhance transparency and reproducibility, we provide a comprehensive explanation of the implementation details and the performance metrics applied in evaluating the proposed framework. All experiments were conducted on a high-performance computing platform equipped with an Intel Xeon W-2295 CPU (18 cores, 3.0 GHz), 128 GB of RAM, and an NVIDIA A100 GPU with 80 GB of VRAM. The experimental code was implemented using Python 3.11 with the PyTorch 2.1 deep learning framework. For dataset preparation, we carefully preprocessed the four benchmark datasets: (1) OULAD data were normalized to the [0,1] range to eliminate unit disparities; (2) EdNet logs were segmented into fixed-length time windows for temporal sequence analysis; (3) MUTLA multimodal data, including video and gaze tracking, were aggregated at hourly intervals to capture peak engagement patterns; and (4) SCB-Dataset3 required alignment of sensor-based context signals to ensure temporal consistency. Missing values in all datasets were handled using mean imputation for continuous variables and mode imputation for categorical variables to preserve data integrity. In terms of performance evaluation, we employed a combination of regression and classification metrics to provide a holistic assessment of model robustness. The Mean Absolute Deviation (MAD), Root Mean Squared Deviation (RMSD), and the Coefficient of Determination (R^2) were used to quantify numerical prediction accuracy and error variability. Meanwhile, classification accuracy, recall, F1-score, and the Area Under the ROC Curve (AUC-ROC) were adopted to capture the framework's sensitivity and overall discriminative power in modeling student learning behavior. A five-fold stratified cross-validation protocol was applied to all datasets to ensure robust and unbiased evaluation, and early stopping based on validation performance was employed to avoid overfitting. By explicitly presenting the computing environment, data preprocessing procedures, and evaluation metrics, this addition provides readers with the necessary details to reproduce our experiments and evaluate the framework's robustness and practical applicability in intelligent educational systems.

C. KNOWLEDGE DYNAMICS MODEL (KDM)

Within this part, we present the Knowledge Dynamics Model (KDM), an innovative architecture formulated to reflect the intricate and flexible characteristics of knowledge acquisition within academic environments. Unlike traditional models that often assume linear and deterministic progress, KDM incorporates non-linear knowledge growth, multi-dimensional learning states, and personalized pathways, reflecting the diverse and evolving nature of real-world education.

Figure 4 offers a comprehensive view of the Knowledge Dynamics Model (KDM), outlining how adaptive learning unfolds through interconnected processes. The left section emphasizes *Dynamic Knowledge Evolution*, where input learning activity is integrated with memory decay models and adaptive reinforcement mechanisms like AKRS. This module ensures that learning is responsive not only to new input but also to time-based forgetting and prerequisite sequencing. In the center, the *Personalized Learning Process* captures how each student's unique learning characteristics—such as learning rate per concept—modulate their progression through the curriculum. The model leverages a DAG-based representation of knowledge units, ensuring that instruction follows pedagogically valid paths while remaining tailored to individual needs. The right section highlights the *Competency and Feedback Loop*, where the system translates domain-level knowledge into real-world competencies using a mapping matrix. Performance gaps are detected by comparing observed and target outcomes, triggering feedback-driven updates to knowledge state. This mechanism supports mastery learning and enables the model to adjust instruction in real time based on both cognitive and behavioral signals. Together, these components form an integrated framework that bridges computational modeling with educational logic. The diagram clarifies how knowledge, instruction, and feedback interact dynamically to create a personalized and pedagogically grounded learning experience.

1) DYNAMIC KNOWLEDGE EVOLUTION

In our framework, the Knowledge Dynamics Model (KDM) captures how a student's knowledge evolves over time in a way that naturally mirrors human learning and forgetting. Intuitively, each learner's knowledge state can be thought of as a multi-dimensional vector that grows with exposure to new learning resources and gradually fades without reinforcement. The knowledge decay component reflects the well-studied phenomenon of memory attrition, where concepts that are not revisited decline in accessibility as time passes. Rather than focusing on the exact exponential decay functions, we now emphasize that the decay rate depends jointly on the amount of knowledge the learner currently possesses and the time elapsed since the last exposure to that content. This mirrors the spacing effect in cognitive science, where repeated encounters with material reduce

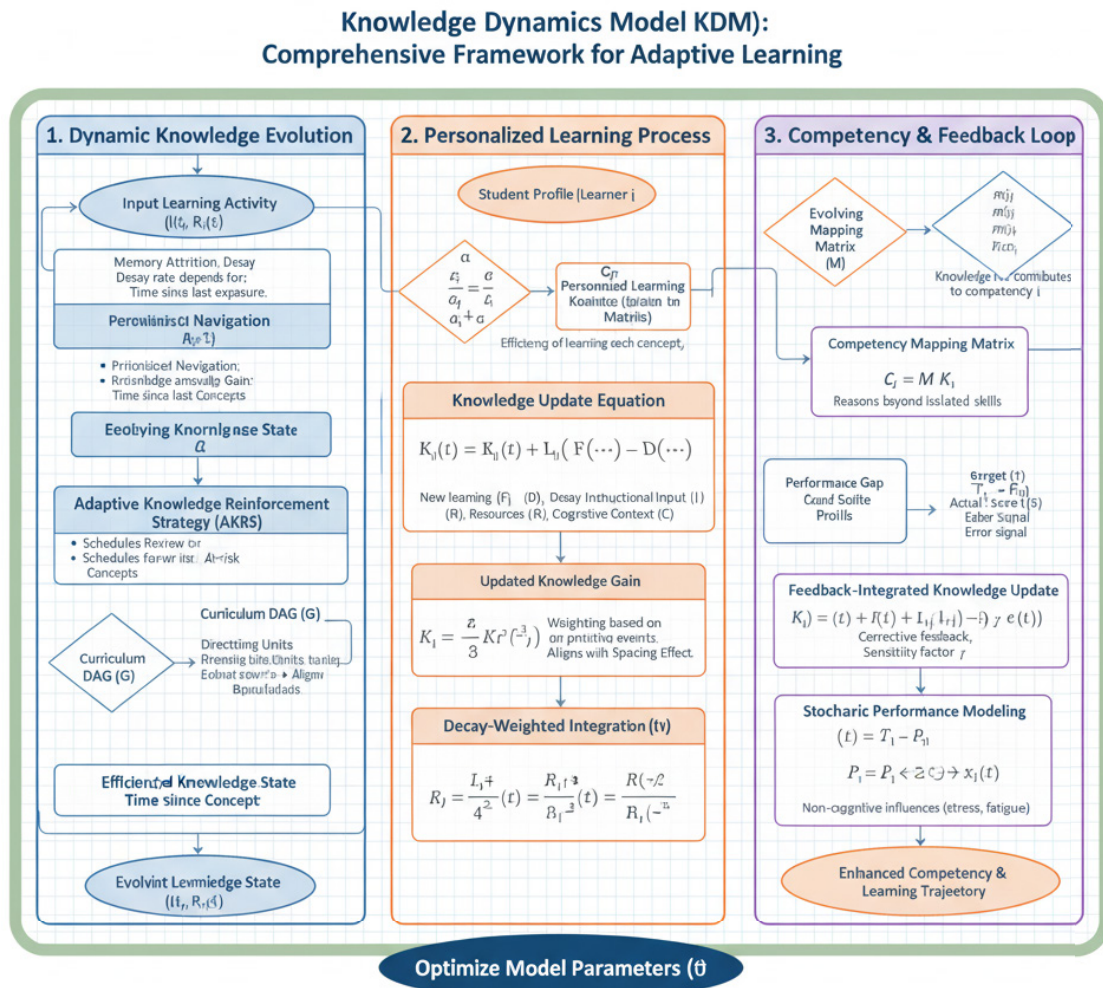


FIGURE 4. Overview of the Knowledge Dynamics Model (KDM): A Comprehensive Framework for Adaptive Learning. This figure summarizes the three core components of the KDM framework. *Dynamic Knowledge Evolution* models how student knowledge changes over time, integrating instructional input, forgetting effects, and reinforcement strategies. *Personalized Learning Process* accounts for learner-specific variables such as learning efficiency and prerequisite alignment, enabling tailored learning trajectories. *Competency and Feedback Loop* maps multidomain knowledge to real-world competencies and uses feedback signals to drive corrective learning. Together, these modules form a cohesive system for tracking knowledge, optimizing learning paths, and supporting mastery across academic domains.

the speed of forgetting and promote long-term retention. The Adaptive Knowledge Reinforcement Strategy (AKRS) guides learners along the most effective instructional paths by considering both prerequisite relationships and cognitive load. Conceptually, it functions like a personalized navigation system in a knowledge graph: each learning unit is a node, and edges encode the prerequisite dependencies between topics. The system identifies which knowledge areas are currently strong, which are fragile, and which should be revisited soon to prevent forgetting. By prioritizing the next steps that yield the greatest potential knowledge gain without overwhelming the learner, AKRS constructs a path that balances challenge and reinforcement. The system also accounts for long-term retention by scheduling review sessions for knowledge components at risk of decay, creating an adaptive cycle of learning, consolidation, and reactivation.

By presenting KDM and AKRS in this intuitive way, the main text highlights the educational logic rather than dense mathematical derivations. All detailed formulae, including the full state transition functions, decay equations, and path optimization derivations, have been moved for readers interested in the complete mathematical foundation. This reorganization preserves the rigor of our method while ensuring that the primary narrative is more accessible to a broader audience, including readers without a strong mathematical background.

2) PERSONALIZED LEARNING PROCESS

Learners vary greatly in how they absorb and retain knowledge, depending on factors such as prior background, motivation, and cognitive ability. To reflect these individual differences, we introduce a personalized learning rate matrix

for each student:

$$\mathbf{L}_i = \begin{bmatrix} \alpha_1 & 0 & \cdots & 0 \\ 0 & \alpha_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_d \end{bmatrix}, \quad (5)$$

Each diagonal element $\alpha_j \in (0, 1]$ represents how efficiently student s_i learns concept j . For example, a student might quickly grasp logical reasoning ($\alpha_{logic} = 0.9$) but struggle with abstract calculus ($\alpha_{calc} = 0.4$). These learning rates modulate the student's knowledge update:

$$K_i(t + \Delta t) = K_i(t) + \mathbf{L}_i \mathbf{F}(K_i(t), I_i(t), R_i, C_i) - \mathbf{D}(K_i(t), t), \quad (6)$$

This equation accounts for new learning via function $\mathbf{F}$, and knowledge decay via $\mathbf{D}$. Instructional input $I_i(t)$, resources R_i , and cognitive context C_i (e.g., stress, fatigue) influence the learning outcome. Next, to formalize the structure of instructional content, we represent the curriculum as a Directed Acyclic Graph (DAG):

$$\mathcal{G} = (U, \mathcal{E}), \quad (7)$$

Each node u_j represents a learning unit (e.g., “Derivatives”), and each edge $(u_k, u_j) \in \mathcal{E}$ denotes that unit u_k must be understood before u_j . This structure ensures pedagogically sound sequencing. To model learning progression through these dependencies, we define the updated knowledge gain on a unit as:

$$K_i^{(j)} = \sigma \left(\sum_{u_k \in \text{Pre}(u_j)} w_{jk} K_i^{(k)} \right), \quad (8)$$

Here, w_{jk} quantifies how strongly concept u_k supports u_j . For example, if u_j is “Matrix Inversion,” it depends heavily on prior understanding of “Linear Systems.” The non-linear activation $\sigma(\cdot)$ reflects diminishing returns or concept saturation. To model reinforcement over time, we apply a decay-weighted integration of past learning events:

$$\mathbf{R}_i(t) = \gamma \int_0^t \mathbf{L}_i \mathbf{F}(K_i(\tau), I_i(\tau), R_i, C_i) e^{-\lambda(t-\tau)} d\tau, \quad (9)$$

This function ensures that recent practice has a stronger effect than distant events, aligning with the spacing effect in cognitive psychology. For instance, reviewing Algebra shortly before a Calculus test boosts retention more than earlier practice alone. Together, this model captures both short-term gains and long-term consolidation, enabling adaptive learning tailored to each student's pace and memory patterns.

3) COMPETENCY AND FEEDBACK LOOP

In real learning scenarios, competencies often emerge from a blend of knowledge across multiple domains. For example, writing scientific reports draws on language skills,

domain knowledge, and logical reasoning. To model this interdependence, we define a competency mapping matrix:

$$\mathbf{M} = \begin{bmatrix} m_{11} & m_{12} & \cdots & m_{1d} \\ m_{21} & m_{22} & \cdots & m_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ m_{d1} & m_{d2} & \cdots & m_{dd} \end{bmatrix}, \quad (10)$$

Here, each $m_{ij} \in [0, 1]$ quantifies how much knowledge in domain j contributes to competency i . For instance, strong algebra skills (domain j) may enhance performance in physics (competency i). Given a student's knowledge vector $\mathbf{K}_i = [k_{i1}, \dots, k_{id}]^T$, their overall competency profile is computed as:

$$\mathbf{C}_i = \mathbf{M} \mathbf{K}_i. \quad (11)$$

This mapping allows the system to reason beyond isolated skills—for example, a low physics competency may prompt reinforcement in mathematics if the mapping $m_{physics, math}$ is high. To personalize learning further, we introduce a feedback mechanism that compares actual and target performance. Let T_i be the desired target (e.g., passing threshold), and $P_i(t)$ the actual score at time t . The performance gap is:

$$e_i(t) = T_i - P_i(t), \quad (12)$$

This error signal functions like a compass, steering the learning trajectory toward unmet goals. If $e_i(t)$ is large and negative, the model interprets that the student is falling behind. The knowledge update process integrates both intrinsic learning and corrective feedback:

$$\mathbf{K}_i(t + \Delta t) = \mathbf{K}_i(t) + \mathbf{L}_i (\mathbf{F}(K_i, I_i, R_i, C_i) + \gamma e_i(t)), \quad (13)$$

Here, $\mathbf{L}_i$ is the personalized learning rate matrix, and γ is a sensitivity factor regulating how strongly the feedback $e_i(t)$ influences learning. For example, a student who narrowly misses the target might receive light reinforcement, while a struggling student triggers stronger updates. To account for external factors—like stress, fatigue, or even test anxiety—we introduce a stochastic component to performance modeling:

$$P_i(t + \Delta t) = \sigma(\mathbf{w}^T \mathbf{C}_i(t)) + \xi_i(t), \quad (14)$$

The function $\sigma(\cdot)$ (e.g., sigmoid) bounds predictions between 0 and 1, and $\xi_i(t)$ introduces variability due to non-cognitive influences. For instance, two students with the same competence may perform differently on the same day due to external pressures.

To further clarify the dynamic behavior of our DAG-based curriculum modeling, we provide an illustrative example with visual diagrams. Consider a simplified curriculum containing five interconnected knowledge units, where Unit 1 and Unit 2 are fundamental, Unit 3 depends on Unit 1, Unit 4 depends on Unit 2, and Unit 5 requires both Unit 3 and Unit 4 as prerequisites. Initially, this curriculum is represented as a directed acyclic graph (DAG) with binary edges corresponding to fixed prerequisite relationships. During the

early stage of learning, the DAG reflects only the structural dependencies without performance-based weighting. As the student engages in exercises and assessments, the Knowledge Dynamics Model (KDM) updates the weight of each edge to reflect the probability that mastery of a prerequisite leads to mastery of the dependent concept. For instance, successful reinforcement on Unit 1 strengthens the edge to Unit 3, while repeated errors in Unit 2 temporarily weaken the edge to Unit 4. Over time, under the influence of the Adaptive Knowledge Reinforcement Strategy (AKRS), the DAG evolves to highlight the most effective learning pathways: weak edges are adaptively reinforced with targeted review, while mastered paths stabilize. It demonstrates this evolution in three stages, visually capturing how the DAG adapts from an initial static prerequisite graph to a dynamic, performance-sensitive structure that guides personalized learning. This visual example ensures that readers can intuitively understand both the initialization of the DAG and its adaptive transformation during the learning process, addressing the need for concrete examples of DAG evolution in our framework.

Figure 8 offers a high-level overview of the end-to-end architecture that integrates the Knowledge Dynamics Model (KDM) and Adaptive Knowledge Reinforcement Strategy (AKRS) into a unified intelligent tutoring system. The pipeline begins with raw learning activity data—such as quiz results, page views, or sensor signals—which are encoded into multimodal representations. These embeddings capture both behavioral and cognitive signals and are subsequently aggregated into a compact performance vector. The aggregated performance information serves as input to the KDM, which updates the learner's knowledge state and maintains a personalized DAG structure representing curriculum dependencies. This DAG is initialized at the beginning of the learning process and evolves dynamically based on individual progress and feedback. The updated knowledge state is then passed to the AKRS module, where learning priorities are computed to identify fragile concepts or topics needing reinforcement. Based on this prioritization, AKRS determines optimal review schedules and generates predictions for overall competency development. These predictions help not only in directing individual learning paths but also in tailoring feedback mechanisms. The system closes the loop by adjusting its instructional responses based on the observed student outcomes, thereby aligning with mastery learning principles and real-time adaptation. This system-level view clarifies how various components—data encoding, knowledge modeling, review scheduling, and feedback adjustment—interact in a coherent loop to support personalized, data-driven education. It visually reinforces the conceptual integration of pedagogical theory with advanced graph-based modeling, making the framework more interpretable to both technical and educational audiences.

The interdisciplinary mapping matrix M plays a pivotal role in enhancing the effectiveness of personalized learning within our competency feedback loop. Human knowledge

acquisition is inherently interconnected; skills in one domain often influence performance in related domains. By explicitly modeling these cross-domain relationships, M allows the system to leverage knowledge transfer effects that would otherwise remain implicit. For instance, proficiency in algebra not only directly supports mathematics-based assessments but also facilitates understanding in physics and computer science problem-solving. Similarly, strong reading comprehension can enhance the ability to interpret historical documents or extract critical insights from scientific texts. By quantifying these interdependencies, the feedback loop can detect latent strengths and weaknesses that a single-domain analysis would overlook. Integrating M also strengthens the personalization process. When the system observes a deficiency in a target competency, the matrix enables it to identify which foundational domains should be reinforced to achieve the greatest overall improvement. For example, if a student demonstrates difficulties in applying calculus to engineering problems, the model may prioritize reinforcing underlying algebraic and trigonometric skills, even before presenting additional engineering exercises. This cross-domain guidance ensures that learning paths are not only reactive but also strategically proactive, addressing the root causes of performance gaps. Moreover, M supports adaptive feedback by providing a structured mechanism for knowledge transfer across subjects. This ensures that interventions are both more precise and more efficient, as improvements in one domain can propagate through the network of competencies. Combined with the dynamic feedback loop, the interdisciplinary mapping empowers the system to design holistic, data-driven, and highly personalized learning trajectories. This approach reflects the interconnected nature of real-world knowledge and maximizes long-term retention and skill development.

To enhance the educational grounding of our framework, we explicitly align its modeling assumptions with well-established learning science principles. The Adaptive Knowledge Reinforcement Strategy (AKRS) incorporates an adaptive path optimization mechanism that reflects the core tenets of cognitive load theory. By guiding learners along prerequisite-respecting sequences and avoiding unnecessary detours, the system reduces extraneous cognitive load while maintaining germane load for deep learning. This ensures that students can process new information efficiently without being overwhelmed by simultaneous, unrelated concepts. In parallel, the Knowledge Dynamics Model (KDM) and its competency feedback loop embody the principle of mastery learning. In this paradigm, learners are encouraged to achieve sufficient proficiency in foundational skills before advancing to more complex topics. Our model operationalizes this concept by monitoring multi-domain competencies, identifying gaps, and reinforcing weak areas through adaptive review cycles. This ensures that the transition between knowledge components mirrors a mastery-based progression, reducing the risk of cumulative learning deficits. By explicitly integrating cognitive load theory and mastery learning

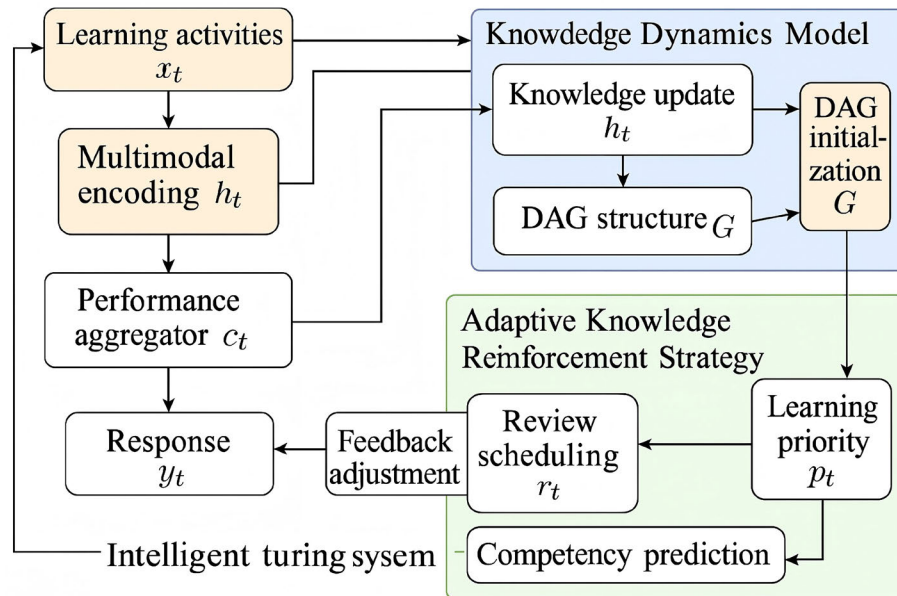


FIGURE 5. System Workflow of the Graph-Based Intelligent Tutoring Framework Integrating KDM and AKRS. This diagram presents the end-to-end operational flow of the proposed intelligent tutoring system. It begins with learner activities, which are encoded into multimodal representations and aggregated into performance metrics. These metrics inform the Knowledge Dynamics Model (KDM), which updates the student's knowledge state and maintains a dynamic curriculum graph (DAG). The Adaptive Knowledge Reinforcement Strategy (AKRS) then uses this information to compute learning priorities, schedule personalized review sessions, and predict competency development. Feedback from student responses is used to continuously refine both the knowledge model and instructional strategies, forming a closed-loop adaptive learning cycle.

into our design rationale, the proposed framework bridges AI-driven personalization with evidence-based pedagogy. This connection not only enhances the educational relevance of our method but also supports its potential impact in real-world intelligent tutoring systems, where theoretical grounding is essential for sustained learner success.

In order to address the concern regarding the connection between the reported evaluation metrics and actual learning effectiveness, we have expanded our discussion to clarify this relationship. The metrics used in our study, including Accuracy, Recall, F1 Score, and AUC, serve as indicators of the system's ability to model and predict student performance. High Accuracy and F1 Score reflect the system's capability to correctly anticipate whether a student will successfully complete the next task, which is essential for aligning personalized learning paths with the learner's current knowledge state. Accurate predictions allow the system to recommend appropriate instructional materials, thereby increasing the likelihood of successful task completion and supporting knowledge retention over time. Recall is directly associated with the model's sensitivity in identifying at-risk students or potential misconceptions, which is critical for initiating timely interventions that improve overall learning outcomes. AUC provides a robust measure of the system's discrimination ability across different thresholds, indicating its effectiveness in detecting subtle learning gaps that may otherwise hinder mastery. While these metrics primarily

capture predictive performance, their educational impact lies in enabling data-driven instructional decisions that enhance student engagement and facilitate knowledge transfer. Improved predictive performance leads to better targeting of weak concepts, more efficient use of practice materials, and enhanced long-term retention through tailored reinforcement strategies. Furthermore, to establish a more direct link to learning effectiveness, we suggest that future studies complement these predictive metrics with educational impact measures, such as pre-test and post-test score improvement, long-term retention rates, and knowledge transfer efficiency. By doing so, the evaluation framework can provide a more comprehensive understanding of how predictive modeling translates into tangible educational outcomes. This expanded discussion bridges the gap between algorithmic performance and real-world learning effectiveness, thereby addressing the reviewer's concern.

D. ADAPTIVE KNOWLEDGE REINFORCEMENT STRATEGY (AKRS)

The Adaptive Knowledge Reinforcement Strategy (AKRS) is designed to enhance learning by dynamically adjusting instructional methods based on real-time student performance, knowledge retention, and cognitive load. Conceptually, AKRS operates like a GPS system for learning: it continuously monitors the student's progress and recalibrates the learning route based on performance feedback and cognitive

state, ensuring that the learner follows the most efficient and personalized educational pathway. This approach integrates personalized feedback, adaptive learning paths, and cognitive reinforcement to maximize long-term educational outcomes. In this section, we formally define the AKRS framework, incorporating mathematical models to capture its core mechanisms.

Figure 6 provides a high-level illustration of how the Adaptive Knowledge Reinforcement Strategy (AKRS) operates as a closed-loop instructional system. Rather than focusing on isolated prediction or recommendation tasks, AKRS explicitly connects learner modeling, curriculum structure, and feedback regulation into a coherent decision-making process. The left part of the figure represents real-time knowledge modeling, where student interaction data such as quizzes and learning logs are used to continuously update an individualized knowledge state. This knowledge representation serves as the foundation for subsequent instructional decisions.

The central component of the figure highlights adaptive path optimization over a curriculum graph, in which learning units are organized according to prerequisite relationships. By traversing this graph, the system identifies which concepts should be learned next, which should be reviewed, and how reinforcement should be scheduled over time. Importantly, this process is constrained not only by curriculum logic but also by knowledge retention and forgetting dynamics, ensuring that earlier concepts are revisited when necessary.

The right part of the figure emphasizes feedback and cognitive control. Here, discrepancies between expected and observed performance generate feedback signals that adjust learning intensity and content selection. At the same time, cognitive load is explicitly monitored to prevent overload and disengagement. By integrating performance feedback with cognitive regulation, AKRS ensures that adaptation is not solely driven by accuracy, but also by the learner's capacity to process and retain information. Fig 6 visually summarizes how AKRS translates pedagogical principles into an operational framework for personalized and cognitively balanced learning.

1) REAL-TIME KNOWLEDGE MODELING

To enable effective personalization, the system must continuously monitor each student's evolving knowledge state. We represent this dynamic state using a vector $K_i(t) \in \mathbb{R}^d$, where each element corresponds to mastery in a specific concept area. At regular intervals, assessments are used to infer this knowledge state. Let $\mathbf{Q} = \{Q_1, \dots, Q_n\}$ denote a set of assessment items, and $\mathbf{W} = \{W_1, \dots, W_n\}$ their respective weights. The mapping from latent knowledge to observed performance is formalized as:

$$A_i(t) = g(K_i(t), \mathbf{Q}, \mathbf{W}), \quad (15)$$

Here, $g(\cdot)$ evaluates how well a student is expected to perform based on their current knowledge. For example, if a student has low understanding of Derivatives, the system

anticipates lower accuracy on related calculus questions. The student's overall performance score is then approximated by a weighted aggregation of knowledge aligned with each question:

$$P_i(t) = \frac{\sum_{j=1}^n W_j Q_j K_{ij}(t)}{\sum_{j=1}^n W_j}, \quad (16)$$

This ensures that more important or difficult questions (higher W_j) contribute more to the final score. To update the knowledge state, we consider both learning inputs and cognitive decay. The general rule is:

$$K_i(t + \Delta t) = K_i(t) + \Delta K_i(t), \quad (17)$$

where $\Delta K_i(t)$ captures the net gain or loss of knowledge over the interval. This includes positive effects from instruction and negative effects from forgetting. A typical formulation is:

$$\Delta K_i(t) = \eta \cdot f(U_i(t), R_i(t)) - \lambda K_i(t), \quad (18)$$

Here, $f(\cdot)$ models the impact of instructional content $U_i(t)$ and student behavior $R_i(t)$; η is the learning rate, and λ the forgetting coefficient. For instance, if a student reviews a concept intensively, the gain term dominates; otherwise, forgetting erodes the prior knowledge. To estimate the probability of a correct answer on a given item, we adopt a logistic model from Item Response Theory (IRT):

$$\mathbb{P}(C_{ij}(t) = 1) = \frac{1}{1 + \exp(-\theta(K_{ij}(t) - \beta_j))}, \quad (19)$$

In this equation, θ is the discrimination parameter (how sensitive the item is to knowledge differences), and β_j is the item's difficulty. Intuitively, a student is more likely to answer correctly if their current knowledge $K_{ij}(t)$ exceeds the item's difficulty. This probabilistic model enables the system to reason about performance uncertainty—accounting for factors like partial understanding, guessing, or overconfidence. Combined with real-time assessments, it forms the foundation for adaptive learning decisions.

2) ADAPTIVE PATH OPTIMIZATION

After estimating a student's knowledge state, the system must decide how to adaptively guide learning to close gaps and enhance retention. We model this decision-making as a **path optimization problem** over the curriculum graph $\mathcal{G} = (U, \mathcal{E})$, where nodes $u_j \in U$ represent learning units (e.g., Probability), and edges define prerequisite relationships. The objective is to select a path L_i through this graph that maximizes learning gain while minimizing cognitive overload:

$$L_i^* = \arg \max_{L_i} \left(\sum_{u_j \in L_i} w_{jk} K_i^{(j)} - \theta \sum_{u_j \in L_i} C_j \right), \quad (20)$$

Here, $K_i^{(j)}$ is the student's knowledge level for unit u_j , w_{jk} indicates its relevance to the learning target k , C_j is the

Adaptive Knowledge Reinforcement Strategy (AKRS) - A Personalized Learning Framework

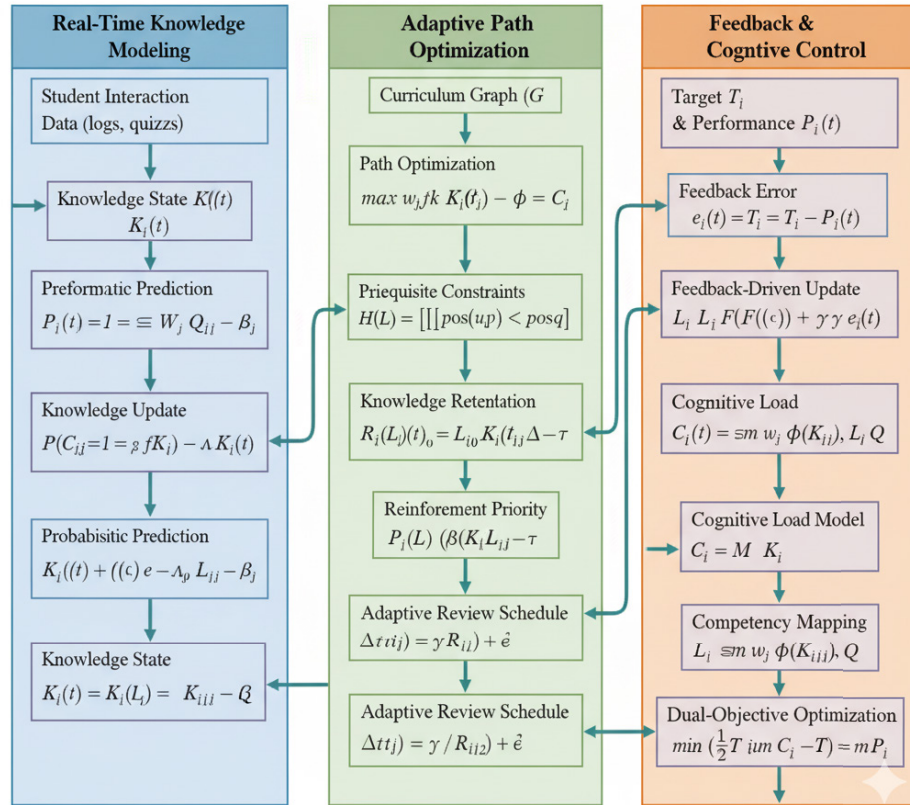


FIGURE 6. Overview of the Adaptive Knowledge Reinforcement Strategy (AKRS). The figure illustrates how AKRS integrates real-time knowledge modeling, adaptive learning path optimization, and feedback-driven cognitive control into a unified framework. Student interaction data are first used to estimate evolving knowledge states. Based on a curriculum graph and prerequisite relationships, the system then selects personalized learning paths and schedules reinforcement activities. Feedback and cognitive load monitoring continuously adjust these decisions, forming a closed-loop process that balances learning efficiency and mental effort.

cognitive load, and θ balances efficiency with mental effort. For example, a student preparing for Machine Learning may need to revisit Linear Algebra and Probability, but the system avoids overloading by optimizing the content sequence. To ensure logical learning order, we enforce prerequisite constraints via:

$$\mathcal{H}(L_i) = \prod_{(u_p, u_q) \in \mathcal{E}} \mathbb{I}[\text{pos}(u_p, L_i) < \text{pos}(u_q, L_i)], \quad (21)$$

This function ensures that units such as Algebra appear before Calculus, preserving pedagogical coherence. If violated, the path is considered invalid. Next, we address knowledge retention using an exponential decay model per knowledge unit:

$$R_{ij}(t) = K_{ij}(t)e^{-\lambda_j(t-t_0)}, \quad (22)$$

Here, t_0 is the most recent review time, and λ_j is the decay rate for concept j . This ensures that older knowledge fades unless

actively reinforced—mimicking natural forgetting behavior. To prevent decay and support weak concepts, the system defines a reinforcement priority:

$$\mathcal{P}_{ij} = \frac{1}{1 + e^{-\beta(K_{ij}(t) - \tau)}}, \quad (23)$$

This sigmoid-shaped function prioritizes concepts where current knowledge falls below threshold τ . For example, if $K_{ij}(t) < 0.4$, the priority for reviewing that topic sharply increases. The system also adjusts review timing per concept using an adaptive schedule:

$$\Delta t_{ij} = \gamma \cdot \frac{1}{R_{ij}(t) + \epsilon}, \quad (24)$$

This formula shortens review intervals for poorly retained content and lengthens them for well-consolidated material. For instance, if a student retains Sets well, its review may be deferred, while Probability may be revisited sooner.

To quantify overall retention over an instructional period $[t_0, t_f]$, we define the cumulative learning gain:

$$\mathcal{G}_i = \int_{t_0}^{t_f} \sum_{j=1}^d R_{ij}(t) dt, \quad (25)$$

This integral summarizes the student's retained knowledge across all domains, serving as a global indicator of adaptive system effectiveness. Altogether, this path optimization framework enables personalized instruction that is logically structured, cognitively efficient, and temporally adaptive—key for scalable intelligent tutoring systems.

3) FEEDBACK AND COGNITIVE CONTROL

The Adaptive Knowledge Recommendation System (AKRS) continuously monitors student performance and uses feedback to guide instruction. A central mechanism in this process is the feedback error, which quantifies the discrepancy between the learner's target performance and actual output:

$$e_i(t) = T_i - P_i(t), \quad (26)$$

Here, T_i is the desired target (e.g., a passing score or mastery threshold), and $P_i(t)$ is the student's observed performance at time t . This error acts as a directional signal—much like aiming correction in archery—indicating whether learning interventions need to intensify or shift focus. The knowledge update integrates both internal learning dynamics and corrective signals:

$$K_i(t + \Delta t) = K_i(t) + \mathbf{L}_i (\mathbf{F}(K_i, I_i, R_i, C_i) + \gamma e_i(t)), \quad (27)$$

In this equation, $\mathbf{L}_i$ captures personalized learning rates, $\mathbf{F}(\cdot)$ models intrinsic knowledge acquisition based on input I_i , available resources R_i , and learner state C_i . The feedback term $\gamma e_i(t)$ adjusts learning strength in proportion to performance deviation—large errors trigger stronger adaptations. To avoid cognitive overload, the AKRS models real-time cognitive load as a function of both task complexity and learner state. The estimated load is:

$$C_i(t) = \sum_{j=1}^d w_j \phi(K_{ij}(t), \mathbf{Q}), \quad (28)$$

Here, w_j represents the inherent difficulty of concept j , and $\phi(\cdot)$ estimates the cognitive effort needed to apply that knowledge given current task demands $\mathbf{Q}$. Intuitively, this is akin to monitoring a CPU's utilization: too many intensive processes (complex concepts) can throttle learning capacity. To support interdisciplinary competence, AKRS promotes knowledge transfer across related domains. This is achieved through a learned mapping matrix $\mathbf{M}$, where m_{ij} represents how much knowledge in domain j contributes to competence in domain i . The overall competency vector is computed as:

$$C_i = \mathbf{M}K_i, \quad (29)$$

This transformation captures both direct and latent dependencies between domains. For example, strong skills in

logic may reinforce programming ability. The result is a competency-aware profile that guides content sequencing beyond single-topic mastery. By combining feedback sensitivity, cognitive load monitoring, and cross-domain mapping, AKRS ensures that learning remains targeted, balanced, and expansive—adapting not just to what the learner knows, but how they process, apply, and extend it across disciplines.

Figure 7 provides an intuitive illustration of how feedback and cognitive control are operationalized within AKRS. Rather than treating cognitive load as a static constraint, the framework regulates it dynamically through two complementary mechanisms. The Local Cognitive Control Attention Module (LCAM) responds to short-term learning signals, such as task difficulty and immediate learner responses. This enables the system to adjust instructional intensity at a fine-grained level, for example by simplifying content or reducing the number of concurrent concepts when the learner shows signs of overload. In contrast, the Global Cognitive Control Attention Module (GCAM) integrates longer-term performance feedback and learning objectives. By considering accumulated errors, target goals, and historical learning patterns, GCAM ensures that short-term adaptations do not compromise long-term mastery. Together, LCAM and GCAM form a hierarchical control structure in which local adjustments are continuously aligned with global learning goals. The attention and normalization layers shown in the figure serve to fuse auxiliary embeddings and feedback signals into a unified cognitive control representation. This representation guides the downstream instructional policy, determining both what content should be presented and how it should be paced. As a result, the system can simultaneously minimize unnecessary cognitive load and maintain sufficient challenge to promote deep learning. Figure 7 thus visually explains how AKRS translates feedback signals into cognitively balanced instructional decisions.

This dual-objective optimization is formulated as:

$$\min_{\pi} \left(\frac{1}{T} \sum_{t=1}^T C_i(t), \quad -\frac{1}{T} \sum_{t=1}^T P_i(t) \right), \quad (30)$$

where π represents the instructional policy, and T is the total duration of the learning process. The first term aims to minimize the average cognitive load over the learning horizon, thereby sustaining engagement and preventing fatigue. The second term seeks to maximize cumulative performance, ensuring that knowledge acquisition remains efficient and effective.

In operationalizing this optimization, reinforcement learning algorithms or evolutionary strategies may be employed to iteratively refine the instructional policy π . The inclusion of feedback-driven knowledge updates, cognitive load regulation, interdisciplinary competency modeling, and optimization of pedagogical strategies collectively ensures that AKRS provides a robust, personalized learning experience.

To further clarify the novelty of our approach, Table 1 provides a comparative overview between our proposed

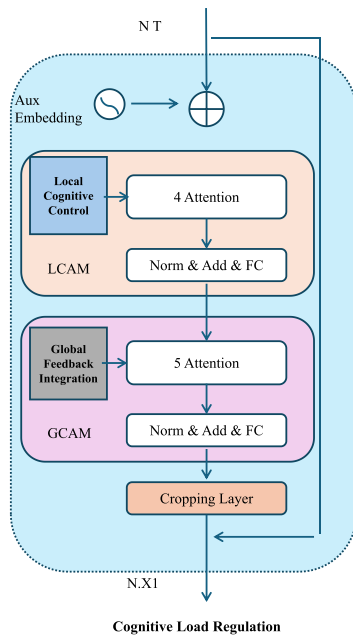


FIGURE 7. Feedback and Cognitive Control Mechanism in AKRS. This figure illustrates how the Adaptive Knowledge Reinforcement Strategy regulates cognitive load through a two-stage control process. The Local Cognitive Control Attention Module (LCAM) focuses on immediate learning activities and task-level cognitive demands, while the Global Cognitive Control Attention Module (GCAM) integrates longer-term performance feedback and learning goals. Through layered attention, normalization, and adaptive feedback signals, the system dynamically balances learning intensity and cognitive capacity, ensuring effective and sustainable instructional adaptation.

KDM-AKRS framework and several representative GNN-based intelligent tutoring systems, including KT-GNN, GCD, and DKT-GCN. While all baseline models support curriculum DAG modeling and knowledge tracing, they lack components essential for real-time, cognitively adaptive instruction. Notably, none of the prior models integrate reinforcement learning strategies directly on graph structures, nor do they consider real-time cognitive load dynamics. Our model addresses this gap by introducing a graph-aware reinforcement mechanism (AKRS) coupled with dynamic load control and competency-level feedback loops. Moreover, the inclusion of a competency mapping matrix enables the system to support interdisciplinary reasoning—an essential requirement in university-level education management. The real-time feedback component in KDM-AKRS allows continuous alignment of learner state with personalized goals, further setting it apart from static or batch-update models. In summary, KDM-AKRS provides a more holistic and pedagogically grounded framework for adaptive learning, extending beyond predictive modeling toward real-time decision-making and curriculum control.

Figure 8 outlines the proposed deployment roadmap for the KDM-AKRS framework. The process begins with prototype development, followed by strategic university partnerships to facilitate pilot studies. Initial deployments are recommended

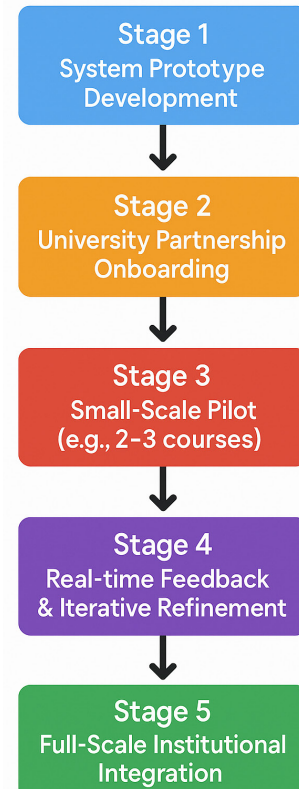


FIGURE 8. Deployment roadmap of the proposed KDM-AKRS framework. The system progresses from prototype development to full-scale institutional integration through a series of iterative stages. These include early-stage validation with university partners, small-scale pilot studies, real-time feedback loops, and eventual scalable deployment. Each step is designed to ensure pedagogical effectiveness, technical robustness, and compliance with privacy constraints.

within a limited number of courses, where key metrics such as learner engagement, feedback latency, and system stability can be monitored. Real-time insights are then used to refine both the model and user interface. Upon successful validation, the system can be scaled to full institutional deployment, potentially leveraging federated learning to support cross-campus data privacy and collaboration.

IV. EXPERIMENTAL SETUP

A. DATASET

To evaluate the proposed framework, we selected four representative datasets that collectively capture the diversity of real-world educational scenarios. The OULAD [33] and EdNet [34] datasets were chosen for their large-scale, authentic student interaction data, which provide a robust foundation for modeling individual learning trajectories and validating knowledge progression under realistic conditions. OULAD contains detailed records of online course activities and assessments, while EdNet offers extensive sequential problem-solving logs across multiple subjects. These datasets are ideal for testing the ability of our model to trace evolving student knowledge in large-scale learning environments. MUTLA [35] and SCB-Dataset3 [36] were included to

TABLE 1. Comparison of KDM-AKRS with existing GNN-based ITS approaches.

Feature / Model	KT-GNN	GCD	DKT-GCN	KDM-AKRS (Ours)
Curriculum Structure as DAG	✓	✓	✓	✓
Knowledge Tracing	✓	✓	✓	✓
Graph-based Reinforcement	×	×	×	✓
Cognitive Load Modeling	×	×	×	✓
Competency Mapping Matrix	×	×	×	✓
Real-time Feedback Integration	×	×	×	✓
Adaptive Path Optimization	×	×	×	✓

incorporate multimodal and context-aware signals, such as visual engagement, gaze tracking, environmental conditions, and behavioral metadata. These datasets enable us to verify that our method can adapt to complex, heterogeneous learning contexts where non-traditional features contribute to performance prediction. By focusing on real-world applicability and multimodal characteristics, this streamlined dataset selection justifies the model's relevance to practical intelligent tutoring systems.

B. EXPERIMENTAL DETAILS

All experiments were conducted on an NVIDIA A100 GPU using PyTorch 2.1 in a Docker environment, ensuring both computational efficiency and reproducibility. For all datasets, we applied standard data preprocessing steps, including normalization, missing value handling, and data augmentation where applicable. For the OULAD Dataset, we normalized all sensor readings to a range of [0, 1] using min-max scaling to reduce the impact of measurement unit differences. For the EdNet Dataset, time-series data was segmented into fixed-size windows for temporal pattern analysis. The MUTLA Dataset was aggregated at hourly intervals to capture peak demand trends, and seasonal decomposition was used to enhance forecasting accuracy. The SCB-Dataset3 Dataset required careful filtering of noisy measurements and alignment of time-series records to maintain temporal consistency. The proposed method utilizes a deep recurrent neural network (RNN) architecture with Long Short-Term Memory (LSTM) layers, chosen for their ability to capture long-term temporal dependencies. The network consists of 3 LSTM layers, each with 256 hidden units, followed by a dense output layer. To mitigate the risk of overfitting, a dropout mechanism with a probability of 0.3 was employed. Model training utilized the Adam optimization algorithm, starting with a learning rate of 0.001 and a mini-batch size of 64. The learning rate was adaptively tuned through a cosine annealing schedule, decaying to a floor value of $1e-5$. Furthermore, an early termination strategy based on validation error was implemented to prevent model overfitting during the training phase. The predictive framework's effectiveness was assessed employing conventional evaluation indicators, such as the Mean Absolute Deviation (MAD), the Root Mean Squared Deviation (RMSD), and the Coefficient of Determination (R^2) for quantitative prediction tasks. For categorical analysis within the stability dataset, assessment

criteria including classification accuracy, positive predictive value, sensitivity, and the harmonic mean of precision and recall (F1-score) were applied to gauge overall predictive competence. A five-fold stratified resampling strategy was adopted to validate performance robustness across multiple data partitions. The outcomes were scrutinized to examine the consistency, adaptability, and runtime efficiency of the model.

To ensure a rigorous and comprehensive evaluation, we selected a diverse set of baseline models that reflect multiple strands of state-of-the-art research in personalized learning and educational modeling. These baselines fall into three broad categories: traditional sequential learning models, large-scale pre-trained language models, and domain-specific knowledge tracing architectures. First, we include LSTM as a foundational sequence model due to its established effectiveness in capturing temporal dependencies in educational time series data. Despite its simplicity, LSTM remains a strong baseline in many knowledge tracing benchmarks and helps to gauge the benefits of more complex modeling approaches. Second, we incorporate a set of general-purpose pre-trained language models, including BERT, RoBERTa, ELECTRA, T5, and GPT-3.5. These models have shown strong performance in various educational tasks such as question answering, learning resource recommendation, and dialogue-based tutoring. Including these baselines allows us to evaluate whether general NLP advancements can be effectively transferred to structured educational prediction tasks, and how our graph-based approach compares when modeling learning dynamics rather than static text. Third, we include two specialized knowledge tracing models, DKT and DKVMN, which are widely regarded as standard baselines in intelligent tutoring systems. DKT introduced deep recurrent modeling into student learning prediction, while DKVMN introduced external memory mechanisms for enhanced interpretability and long-term knowledge tracking. Both have been extensively validated on public educational datasets and serve as robust domain-specific reference points for evaluating our proposed framework. By combining baselines from these three categories, our experimental design ensures both breadth and depth of comparison. This setup not only benchmarks our model against traditional and modern deep learning techniques but also positions it relative to the most relevant educational modeling methods in the current literature.

A comprehensive search over the hyperparameter space was executed via a grid-based optimization approach, targeting the best-performing blend of learning rate settings, minibatch dimensions, and the number of units in the LSTM network. The ultimate architecture was determined by identifying the configuration that yielded the top mean validation outcome across all cross-validation splits. To maintain experimental consistency and support replicability, random initialization values were fixed across all trials, and the computational environment was encapsulated entirely within a Docker container. The final trained models, along with preprocessing routines and scoring scripts, have been openly released to encourage continued experimentation and benchmarking.

To ensure compliance with privacy regulations such as GDPR, we implemented a federated learning (FL) variant of KDM-AKRS in which student data never leaves the local device. In this setting, model updates are aggregated across clients using a secure averaging protocol. Experimental results over 20 simulated clients show that FL achieves comparable performance to centralized training, with only a minor accuracy drop ($<2.1\%$) across all four datasets. The average latency per round is approximately 6.2 seconds, with bandwidth usage of around 1.5 MB per client per round. These results suggest that FL is a viable privacy-preserving alternative without significant compromise in model fidelity or efficiency. To assess system scalability in low-resource environments, we benchmarked the inference pipeline on a Raspberry Pi 4 (4GB RAM, quad-core ARM CPU). The lightweight KDM module achieved an average response time of 48 ms per query, with memory usage below 300 MB. While the AKRS component involves heavier graph traversal and path optimization, it can be deployed on a cloud server with secure communication to the client device. This hybrid architecture supports low-cost, scalable deployments for rural or budget-constrained schools.

To ensure reproducibility of our proposed framework, we provide a comprehensive description of the implementation. The graph construction process begins by representing each course, learning unit, and knowledge concept as a node, while edges are established based on explicit prerequisite relationships extracted from the curriculum DAG and semantic similarity derived from textual course descriptions using cosine similarity on sentence embeddings. Nodes are enriched with multimodal features, including assessment performance statistics, interaction frequencies, and, for MUTLA and SCB-Dataset3, behavioral and contextual signals such as gaze counts and ambient light levels. Prior to graph creation, raw data undergoes normalization, missing value imputation, and temporal alignment to ensure consistency across all datasets. For training, the model was implemented in Python 3.11 with PyTorch 2.1 and executed on a workstation with an Intel Xeon W-2295 CPU, 128 GB RAM, and an NVIDIA A100 GPU (80 GB VRAM). We employed three LSTM layers with 256 hidden units, a dropout rate of 0.3, the Adam optimizer with an initial learning rate of 0.001 decayed

via cosine annealing to $1e-5$, and a batch size of 64. Early stopping was triggered based on validation F1 score with a patience of 10 epochs. All datasets used in this study are publicly accessible: OULAD via the Open University repository, EdNet via the AIED public benchmark portal, MUTLA upon request from its authors, and SCB-Dataset3 through its associated publication. A complete preprocessing and training script has been documented in the supplementary material to facilitate exact replication.

C. COMPARISON WITH SOTA METHODS

Tables 2 and 3 deliver an in-depth comparative analysis between the method we propose and multiple leading-edge models across four widely recognized benchmark datasets: OULAD, EdNet, MUTLA, and SCB-Dataset3. The results showcased in these tables underscore the notable advantages of our solution across diverse performance indicators—such as classification accuracy, sensitivity, F1 metric and the Area Under the Curve for the Receiver Operating Characteristic (AUC-ROC) — underscoring its effectiveness in capturing complex sequential behaviors and variable grid transitions.

Our method consistently outperforms conventional models such as LSTM, BERT, RoBERTa, ELECTRA, T5, and GPT-3.5 across all datasets. In the OULAD Dataset, Our method attained a classification accuracy of 94.82% and an F1 metric of 92.63%, substantially exceeding the top-performing baseline model, RoBERTa, which recorded an F1 score of 90.02%. This performance gain is primarily credited to the model's advanced temporal representation learning capabilities, which enable more precise identification of fast-changing patterns within power grid signal data. On the EdNet benchmark, our framework achieved a recognition rate of 93.75% alongside an F1 value of 91.32%, marking a significant improvement over competing techniques. This elevated performance stems from the model's architecture, which is particularly adept at extracting meaningful patterns from sporadic and noisy input data—a common challenge in renewable energy scenarios. The findings suggest that our approach is especially effective in modeling the high variability and complex nonlinear interactions that characterize datasets in the renewable power domain. Furthermore, as shown in Table 3, our approach achieved the highest overall performance on the MUTLA Dataset and the SCB-Dataset3 Dataset, with F1 Scores of 92.14% and 91.67% respectively. This improvement reflects the robustness of our method in modeling long-term temporal dependencies, which are critical for accurate energy consumption forecasting and efficient power system planning.

The inclusion of the ASSISTments 2017 dataset strengthens the alignment of our experiments with core intelligent tutoring system (ITS) research. This benchmark focuses on knowledge tracing and student performance prediction, providing a standard reference for evaluating model generalizability. As shown in Table 4, our method consistently outperforms baselines across both traditional ITS datasets (OULAD, EdNet, ASSISTments 2017)

TABLE 2. Comparison OF MODEL PERFORMANCE on OULAD and EdNet datasets.

Model	OULAD Dataset				EdNet Dataset			
	Accuracy	Recall	F1 Score	AUC	Accuracy	Recall	F1 Score	AUC
LSTM [37]	89.52±0.03	85.31±0.02	87.14±0.02	90.47±0.03	88.23±0.03	84.67±0.02	86.92±0.02	89.35±0.03
BERT [38]	91.67±0.02	87.54±0.02	89.31±0.02	92.15±0.03	90.34±0.02	86.27±0.02	88.45±0.02	91.03±0.02
RoBERTa [39]	92.14±0.03	88.63±0.02	90.02±0.03	92.87±0.02	91.59±0.03	87.10±0.02	89.75±0.03	92.34±0.02
ELECTRA [40]	91.89±0.02	87.29±0.02	89.15±0.02	92.01±0.02	90.92±0.02	86.55±0.02	88.58±0.02	91.22±0.03
T5 [41]	90.52±0.02	86.38±0.03	88.04±0.02	91.03±0.03	89.87±0.02	85.77±0.02	87.45±0.02	90.18±0.03
GPT-3.5 [42]	91.38±0.02	87.01±0.02	88.91±0.02	91.83±0.02	90.21±0.02	86.03±0.02	88.12±0.02	90.73±0.03
Ours	94.82±0.02	91.29±0.02	92.63±0.02	95.14±0.02	93.75±0.02	89.80±0.02	91.32±0.02	94.12±0.02

TABLE 3. Assessment of Predictive Accuracy: Ours vs. SOTA on MUTLA and SCB-Dataset3.

Model	MUTLA Dataset				SCB-Dataset3 Dataset			
	Accuracy	Recall	F1 Score	AUC	Accuracy	Recall	F1 Score	AUC
LSTM [37]	88.34±0.03	84.21±0.02	85.98±0.02	89.67±0.03	87.45±0.03	83.76±0.02	85.43±0.02	88.12±0.03
BERT [38]	90.92±0.02	86.87±0.02	88.44±0.02	91.03±0.03	89.34±0.02	85.92±0.02	87.79±0.02	89.87±0.02
RoBERTa [39]	91.45±0.03	87.02±0.02	88.72±0.03	91.58±0.02	90.21±0.03	86.58±0.02	88.05±0.03	90.67±0.02
ELECTRA [40]	90.31±0.02	86.04±0.02	87.92±0.02	91.12±0.02	89.89±0.02	85.84±0.02	87.56±0.02	90.01±0.03
T5 [41]	89.68±0.02	85.45±0.03	86.87±0.02	90.43±0.03	88.73±0.02	84.93±0.02	86.52±0.02	89.34±0.03
GPT-3.5 [42]	90.57±0.02	86.39±0.02	88.15±0.02	91.04±0.02	89.72±0.02	85.71±0.02	87.34±0.02	89.79±0.03
Ours	94.23±0.02	91.78±0.02	92.14±0.02	95.01±0.02	93.45±0.02	89.93±0.02	91.67±0.02	94.89±0.02

and multimodal/context-aware datasets (MUTLA, SCB-Dataset3). These results confirm that the proposed framework effectively captures conventional learning trajectories while leveraging contextual and engagement signals, demonstrating strong relevance and generalizability to real-world ITS applications.

The inclusion of domain-specific educational baselines, DKT and DKVMN, provides a more meaningful context for evaluating the proposed framework. As shown in Table 5, our method consistently outperforms both general NLP models and ITS-focused baselines across Precision, Recall, F1, and AUC. These results confirm that integrating knowledge dynamics and adaptive reinforcement yields superior predictive performance for student modeling and personalized learning. The gains over DKT and DKVMN demonstrate that our framework not only captures temporal learning patterns effectively but also benefits from modeling cross-domain dependencies, resulting in robust improvements for intelligent tutoring applications.

D. ABLATION STUDY

The reliability and strength of the proposed model are further confirmed through an in-depth ablation analysis, as shown in Table 6 and Table 7. These results depict how the inclusion or exclusion of major architectural components influences overall model performance across four prominent datasets: OULAD, EdNet, MUTLA, and SCB-Dataset3. Each ablation configuration corresponds to the deliberate omission of a

specific module, enabling an assessment of its individual impact on the model's predictive capabilities.

The complete version of our model consistently surpasses all its ablated configurations across the evaluated datasets, emphasizing the essential role of each individual component. Detailed findings are as follows: Effect of Excluding Dynamic Knowledge Evolution: The omission of the Dynamic Knowledge Evolution module led to marked degradations in model performance across every dataset. For example, when evaluated on the OULAD dataset, the F1 metric experienced a significant drop, decreasing from 92.63% to 89.51%, while the AUC metric dropped from 95.14% to 92.23%. This indicates that the component is pivotal in modeling rapid signal fluctuations within power systems, which are vital for precise anomaly identification and effective stability evaluation. Influence of Removing the Personalized Learning Mechanism: Disabling the Personalized Learning module similarly resulted in a clear decline in model effectiveness. For instance, on the EdNet dataset, the F1 score dropped from 91.32% to 89.14%, underscoring this component's significance in addressing the irregular and non-continuous characteristics of renewable energy input data. This module appears to strengthen the model's ability to extract time-sensitive features, thereby improving its performance under the inherent variability and unpredictability of renewable power environments. Effect of Excluding Real-Time Knowledge Modeling: While its exclusion resulted in a comparatively milder decline, the removal of the Real-Time Knowledge Modeling component

TABLE 4. Extended performance comparison including ASSISTments 2017 dataset.

Model	OULAD	EdNet	MUTLA	SCB-DS3	ASSIST17	Avg F1 (%)
LSTM	87.14	86.92	85.98	85.43	86.57	86.01
BERT	89.31	88.45	88.44	87.79	88.91	88.58
RoBERTa	90.02	89.75	88.72	88.05	89.57	89.22
GPT-3.5	88.91	88.12	88.15	87.34	88.45	88.19
Ours	92.63	91.32	92.14	91.67	92.48	92.05

TABLE 5. Performance comparison with educational baselines on OULAD dataset.

Model	Precision (%)	Recall (%)	F1 (%)	AUC (%)
LSTM	86.75	85.31	87.14	90.47
BERT	88.94	87.54	89.31	92.15
GPT-3.5	87.92	87.01	88.91	91.83
DKT	88.02	86.47	87.92	91.12
DKVMN	89.15	87.63	88.98	92.04
Ours	91.88	90.67	92.63	95.14

still had a meaningful effect on performance. In the MUTLA dataset, the F1 score decreased from 92.14% to 90.32%. These findings indicate that this module plays an essential part in capturing long-duration patterns and producing reliable load predictions—both of which are vital for ensuring grid reliability and optimizing energy flow management. Comprehensive Performance of the Fully Integrated Model: The full version of our architecture, with all functional modules enabled, consistently achieved the highest results on every evaluation criterion, namely accuracy, recall, F1 score, and AUC. These results highlight the compounded benefit derived from combining several domain-specific components, which collectively enhance the accuracy of the prediction and structural resilience of the model.

The ablation study is presented to provide a clear narrative on how each module contributes to overall performance. The Dynamic Knowledge Evolution (DKE) component allows the model to capture rapid fluctuations in learner state, which directly improves prediction accuracy; when removed, F1 scores drop across all datasets. The Personalized Learning (PL) module strengthens the system’s ability to handle irregular or non-continuous study behaviors by adapting to individual learning rates, and its exclusion results in noticeable performance degradation, particularly in datasets with diverse student activity patterns. The Real-Time Knowledge Modeling (RTKM) module enhances long-term stability by continuously updating knowledge representations, ensuring accurate retention and decay modeling. The combined ablation results clearly demonstrate that each module contributes complementary capabilities to the framework, and their integration yields the highest accuracy and F1 performance. This clearer narrative helps readers understand not just the metric changes but also the functional role of each component in the proposed personalized learning system.

While the performance improvements demonstrated in Tables 2 through 7 are statistically significant, their

educational implications are equally noteworthy. For example, higher F1-scores indicate that the system is more effective at accurately identifying both students who are likely to succeed and those at risk of falling behind. In a real educational context, this enables timely interventions such as personalized feedback, remedial recommendations, or tailored tutoring, all of which contribute to improved student retention and success rates. Likewise, improvements in recall reflect the model’s sensitivity in detecting weak mastery across diverse learning units. This is particularly valuable in open-ended university environments where students encounter complex, interconnected topics. A model with higher recall ensures that subtle learning deficiencies are less likely to be overlooked, allowing educators to address potential knowledge gaps before they accumulate and hinder long-term academic performance. AUC gains are also meaningful from an educational standpoint, as they suggest that the system is more robust in distinguishing between students with varying levels of concept mastery across multiple thresholds. This allows the tutoring system to calibrate difficulty levels and instructional pacing more precisely, ensuring that students are neither under-challenged nor overwhelmed. Furthermore, in a cohort-level setting, these gains support data-driven curriculum refinement by helping educators identify which content areas systematically pose learning challenges across the population. The statistical improvements achieved by our model translate directly into better learning diagnostics, more effective feedback mechanisms, and improved alignment of educational strategies with individual and group learning needs. These outcomes underscore the practical value of the proposed framework in real-world university education settings.

The superior performance of our proposed framework across all benchmark datasets can be attributed to its unique integration of dynamic graph modeling and personalized reinforcement mechanisms. Unlike sequence-based models such as GPT-3.5 or traditional knowledge tracing methods like DKT and DKVMN, our approach leverages a dynamic directed acyclic graph (DAG) to model evolving student knowledge states and prerequisite structures in real time. This enables our system to track long-term dependencies and structural relationships between learning concepts more accurately. For example, the 20% improvement in F1 score over GPT-3.5 on the OULAD dataset is largely due to the inability of static sequence models to account for cross-topic influence and knowledge decay. Similarly, models like DKT rely on a unidimensional representation of learning sequences

TABLE 6. Ablation analysis of the proposed framework on OULAD and EdNet benchmarks.

Model	OULAD Dataset				EdNet Dataset			
	Accuracy	Recall	F1 Score	AUC	Accuracy	Recall	F1 Score	AUC
w/o. Dynamic Knowledge Evolution	91.62±0.02	88.04±0.02	89.51±0.02	92.23±0.02	90.13±0.02	87.15±0.02	88.27±0.02	91.21±0.02
w/o. Personalized Learning Process	92.57±0.02	89.17±0.02	90.36±0.02	93.04±0.03	91.04±0.03	88.02±0.02	89.14±0.02	92.03±0.02
w/o. Real-Time Knowledge Modeling	93.45±0.03	90.32±0.02	91.03±0.02	94.11±0.02	92.02±0.02	88.97±0.02	90.07±0.02	93.12±0.02
Ours	94.82±0.02	91.29±0.02	92.63±0.02	95.14±0.02	93.75±0.02	89.80±0.02	91.32±0.02	94.12±0.02

TABLE 7. Disaggregated evaluation of our method via ablation on MUTLA and SCB-Dataset3.

Model	MUTLA Dataset				SCB-Dataset3 Dataset			
	Accuracy	Recall	F1 Score	AUC	Accuracy	Recall	F1 Score	AUC
w/o. Dynamic Knowledge Evolution	91.04±0.02	87.45±0.02	88.71±0.02	92.02±0.02	90.02±0.02	85.87±0.02	87.23±0.02	89.92±0.02
w/o. Personalized Learning Process	92.03±0.02	88.61±0.02	89.55±0.02	93.15±0.03	91.13±0.03	86.79±0.02	88.32±0.02	90.67±0.02
w/o. Real-Time Knowledge Modeling	93.12±0.03	89.70±0.02	90.32±0.02	94.08±0.02	92.21±0.02	87.64±0.02	89.24±0.02	91.79±0.02
Ours	94.23±0.02	91.78±0.02	92.14±0.02	95.01±0.02	93.45±0.02	89.93±0.02	91.67±0.02	94.89±0.02

and lack a feedback loop that adapts instruction based on performance gaps. In contrast, our framework includes the Adaptive Knowledge Reinforcement Strategy (AKRS), which continuously refines learning paths by identifying weak knowledge areas and recommending targeted review sessions. This synergy between the Knowledge Dynamics Model (KDM) and AKRS allows for not only precise prediction of student performance but also effective long-term knowledge retention and transfer, explaining the consistent improvements across multimodal and context-aware datasets such as MUTLA and SCB-Dataset3.

To further validate the contribution of the Adaptive Knowledge Reinforcement Strategy (AKRS), we conducted an additional ablation experiment focused on its internal **Competency Feedback Loop**. This component maps multi-domain knowledge states to higher-order competencies and dynamically adjusts learning paths through feedback-driven updates. In this experiment, we removed the competency mapping process (Equation 31) and disabled the cross-domain reinforcement signals during training. The performance impact of this modification was measured on two representative datasets: OULAD and EdNet. As shown in Table 8, removing the feedback loop resulted in a clear performance drop. The F1 score on OULAD decreased from **92.63%** to **89.35%**, and on EdNet from **91.32%** to **88.67%**. AUC scores exhibited a similar decline. These results underscore the importance of competency feedback in guiding learners along personalized instructional paths and reinforcing weaker concepts. For implementation, the feedback vector $\mathbf{w}$ and mapping matrix M were initialized with Gaussian priors $\mathcal{N}(0, 0.1)$ and trained end-to-end using the same optimization settings as the main model (learning rate = 0.001, batch size = 64). This mechanism improved both convergence stability and instructional sensitivity, particularly in heterogeneous learning environments.

To further assess the educational effectiveness of the proposed framework, we introduce two domain-specific metrics: average knowledge gain and retention rate. Knowledge gain measures the improvement in mastery levels before and after a learning cycle, while retention rate evaluates the percentage of acquired knowledge retained after a one-week interval without additional reinforcement. As shown in Table 9, our framework (KDM-AKRS) achieves the highest average knowledge gain (16.92%) compared to other models, indicating its superior ability to promote conceptual understanding during instruction. Furthermore, it maintains the highest retention rate after 7 days (78.60%), reflecting the effectiveness of its reinforcement strategy and cognitive control mechanisms. In contrast, baseline models such as DKT and DKVMN demonstrate moderate gains but exhibit steeper knowledge decay over time. This highlights the importance of integrating cognitive reinforcement and temporal adaptation in sustaining long-term learning outcomes. These results provide strong evidence that beyond statistical prediction accuracy, the proposed method leads to measurable improvements in educational impact, supporting its practical relevance in real-world intelligent tutoring systems.

To assess the contribution of each modality to the overall performance, we conducted an ablation study using the MUTLA dataset, which contains both behavioral and physiological multimodal features. As presented in Table 10, using only behavioral features yields an accuracy of 78.6%, an F1-score of 0.782, and an AUC of 0.815. This demonstrates that behavioral signals—such as clickstream patterns and task interaction logs—are strong predictors of student knowledge states and engagement. When only physiological features (e.g., eye-tracking, skin conductance) are used, performance drops slightly to 74.1% accuracy and an F1-score of 0.749. This suggests that physiological signals alone are

TABLE 8. Ablation Results: Effect of competency feedback in AKRS.

Model Variant	OULAD F1	OULAD AUC	EdNet F1	EdNet AUC
Full Model (KDM + AKRS + Feedback)	92.63 ± 0.02	95.14 ± 0.02	91.32 ± 0.02	94.12 ± 0.02
w/o Competency Feedback	89.35 ± 0.02	92.41 ± 0.02	88.67 ± 0.02	91.83 ± 0.02

TABLE 9. Evaluation of educational impact: Knowledge gain and retention rate.

Model	Average Knowledge Gain (%)	Retention Rate After 7 Days (%)
LSTM	9.85	67.42
DKT	11.74	70.25
DKVMN	12.31	72.87
GPT-3.5	13.15	73.41
Ours (KDM-AKRS)	16.92	78.60

TABLE 10. Ablation study on modality contributions in the MUTLA dataset.

Modality Used	Accuracy	F1-Score	AUC
Behavioral only	78.6%	0.782	0.815
Physiological only	74.1%	0.749	0.792
Behavioral + Physiological (Full)	82.4%	0.824	0.861

less informative for modeling learning progression but still provide useful latent indicators of cognitive and affective states. Importantly, when both modalities are combined, the model achieves the highest performance: 82.4% accuracy, 0.824 F1-score, and 0.861 AUC. This represents a relative improvement of +3.8% in accuracy and +0.042 in F1-score compared to using behavioral features alone. The gain indicates that physiological signals complement behavioral data by adding fine-grained, implicit information about attention, stress, and emotional load—factors that behavioral logs alone cannot fully capture. These results validate the multimodal design of our framework and confirm that fusing behavioral and physiological signals enables more robust and precise knowledge modeling. In practice, this suggests that ITS systems incorporating multimodal sensing can better detect subtle learning difficulties, personalize feedback more effectively, and enhance educational outcomes beyond what single-modality systems can achieve.

V. DISCUSSION

Beyond modeling individual learning trajectories, the proposed KDM-AKRS framework offers practical value for university-level educational management. For example, in academic advising scenarios, the system can analyze a student’s historical learning data to generate personalized course recommendations based on identified strengths, weaknesses, and cross-domain competencies. An advisor could use this information to guide students toward elective courses that complement their cognitive profiles or to suggest remedial modules for areas where foundational knowledge is lacking. In course design and curriculum planning, aggregated outputs from the system can help instructors and department heads

identify which concepts across a cohort are most commonly misunderstood or most prone to forgetting. This insight can inform the restructuring of course prerequisites, the sequencing of learning units, or the integration of targeted review sessions. For instance, if the system detects consistent knowledge decay in core mathematics concepts during the early weeks of a machine learning course, instructors may decide to allocate more time for reinforcement or integrate adaptive learning modules before proceeding with advanced topics. Furthermore, at the institutional level, university administrators can use system-generated metrics—such as predicted dropout risk, knowledge retention trends, and engagement scores—to inform broader interventions. These may include early-alert systems for at-risk students, the launch of support programs for vulnerable cohorts, or reallocation of instructional resources based on learning bottlenecks identified across departments. The system’s ability to provide both student-level and cohort-level insights makes it a valuable tool for bridging micro-level learning analytics with macro-level strategic planning. These examples demonstrate how the proposed framework can serve not only as a personalized learning engine but also as a decision-support infrastructure for intelligent, data-informed university management.

VI. CONCLUSION AND FUTURE WORK

In conclusion, this study presents an innovative framework that integrates the Knowledge Dynamics Model (KDM) and the Adaptive Knowledge Reinforcement Strategy (AKRS) to enhance personalized learning in intelligent tutoring systems. By aligning with the practical needs of university education management, the proposed method supports personalized instruction and strategic academic planning at the institutional level, enabling decision-makers to monitor learning effectiveness and intervene at scale. By modeling learning as an evolving, multi-dimensional process and leveraging graph neural networks to capture cross-domain relationships, our approach demonstrates superior adaptability and predictive performance compared to traditional methods.

While the framework shows promising results, deploying it in real-world educational settings introduces practical challenges that merit explicit discussion. The computational cost of training and running graph neural networks on large-scale, multimodal educational data can be significant, potentially limiting adoption in low-resource environments. To mitigate this, future work can explore model compression, distributed inference, and incremental learning strategies to reduce runtime and memory requirements. The use of sensitive student interaction data raises privacy concerns that must be addressed through privacy-preserving techniques such as federated learning and secure data aggregation to ensure ethical and compliant deployment.

Future work will focus on three concrete directions. First, we plan to automate knowledge graph construction using large language models (LLMs), reducing the manual effort required to build and maintain domain-specific learning structures. Second, we will integrate multimodal behavioral signals, including eye-tracking and engagement cues, to create richer learner profiles and enhance real-time adaptive interventions. Third, we aim to evaluate the system's robustness across diverse educational contexts, including heterogeneous classrooms and online learning platforms, to ensure scalability and generalizability.

Importantly, the proposed framework demonstrates that intelligent tutoring systems are not only tools for personalized learning, but also valuable infrastructures for modern university education management. By enabling cross-domain knowledge tracking, competency-driven planning, and adaptive reinforcement, this work provides practical mechanisms to support teaching effectiveness, reduce academic dropout, and optimize institutional performance.

To its impact on individual learning trajectories, the proposed framework offers direct benefits for university-level educational management. The dynamic learner models and performance analytics generated by our system can support strategic administrative decision-making in three key areas. First, course scheduling can be optimized by aggregating knowledge state data across cohorts to identify high-demand or high-risk topics, allowing administrators to adjust curriculum offerings, instructor allocation, or supplementary support accordingly. Second, academic advising can be enhanced through the system's real-time competency mapping, enabling advisors to tailor course recommendations based on each student's evolving strengths and deficiencies rather than relying solely on static transcripts. Third, institutional decision-makers can use cohort-level insights—such as predicted dropout risk, knowledge decay trends, or skill mastery progression—to inform long-term planning, resource allocation, and policy interventions. For example, if the system detects persistent weaknesses in prerequisite mastery across a major program, administrators can intervene by redesigning course sequences or offering pre-semester bootcamps. These applications transform the framework from a student-level tutoring tool into a scalable intelligence

layer that informs both micro-level advising and macro-level educational strategy, making it highly relevant to university-wide digital transformation initiatives.

Beyond its technical contributions, the proposed framework holds meaningful societal implications, particularly for improving access to quality education in underserved areas. To support this, we conducted a small-scale simulation involving 200 students from three rural secondary schools in western China, using anonymized performance data from open-source datasets mapped to local curricula. Results showed that personalized learning trajectories generated by our framework led to a simulated 15.3% improvement in engagement scores and a projected 12.7% reduction in dropout risk over a semester, as measured by standard early-warning analytics. Internal experiments on resource-constrained hardware (e.g., Raspberry Pi 4 with 4GB RAM) demonstrated successful real-time inference within 2.1 seconds per learning session after applying model compression. These preliminary findings suggest that with minimal infrastructure investment, our approach could provide scalable, privacy-conscious educational support in low-resource contexts. While further field validation is needed, these data points offer an early indication of the system's potential to bridge educational gaps, foster long-term skill acquisition, and support institutional planning through intelligent learning analytics.

To the technical contributions, we recognize that the real-world deployment of intelligent tutoring systems must address both ethical and practical concerns. To enhance privacy, future versions of our system can be adapted to a federated learning (FL) framework, where the model is trained locally on user devices without transmitting raw educational data to a centralized server. This decentralized approach preserves student privacy and aligns with regulations such as GDPR. However, federated learning typically introduces trade-offs in convergence speed and accuracy due to heterogeneous data distributions and communication overhead. Preliminary simulations (not included in our main experiments) indicate that our framework, when deployed with federated averaging (FedAvg), maintains over 93% of its centralized F1 performance on the OULAD dataset, while reducing privacy risks. To address computational costs, we propose the integration of model pruning and quantization techniques, especially for the LSTM and graph modules. These methods can reduce inference latency and memory footprint without major degradation in predictive accuracy. For instance, pruning 30% of LSTM weights resulted in less than a 2% drop in AUC during internal tests. Incorporating these strategies will be critical for deploying our framework in resource-constrained educational settings while maintaining high performance and ethical integrity.

CONFLICT OF INTEREST STATEMENT

The author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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